

DATA ANALYST INTERN

Prerequisite Learning Guide & Program Overview

Fintech Division | Analytics & Insights Team

Program	Data Analyst Internship — Cohort 2025
Domain	Fintech Financial Analytics Data Engineering
Duration	2 Weeks (Week 1: Learning Week 2: Evaluation)
Start Date	Immediate — Program Active
Document	Prerequisite Skills & Timeline Guide v1.0

This document outlines everything you need to learn, complete, and demonstrate during your internship. Read it carefully on Day 1.

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01 PROGRAM OVERVIEW & TIMELINE

Welcome to the Data Analyst Internship program. This is a structured 2-week intensive designed to rapidly upskill you in data analytics fundamentals and apply those skills to real Fintech company projects. Your performance across the prerequisite test and capstone project will directly determine the ongoing company projects you are assigned.

Phase	Timeline	Activity	Outcome
Week 1 Day 1-2	Days 1–2	Orientation, prerequisite skill learning begins. Study Python, SQL, Pandas, NumPy, Flask, Fintech basics.	Learning initiated
Week 1 Day 3-5	Days 3–5	Continue skill building. Capstone project assigned (Day 3 or 4). Simultaneous: upskill + begin project work.	Capstone assigned
Week 2 Day 6-7	Days 6–7	Prerequisite Test conducted online. Test covers all technical + Fintech topics.	Test evaluated
Week 2 Day 8-10	Days 8–10	Capstone Project submission deadline. Evaluation by mentor team.	Project submitted
Post Week 2	Day 11+	Based on test + capstone performance, major ongoing company projects assigned.	Projects assigned

02 KEY RULES & EXPECTATIONS

1. You must complete all prerequisite skills learning in Week 1. No extensions.
2. The Capstone Project will be assigned between Day 3–5. You **MUST** start immediately.
3. The Online Prerequisite Test covers all topics listed in this document. Prepare thoroughly.
4. Your Capstone Project must be submitted by the end of Week 2. Late submissions are penalized.
5. Project assignment post-internship is **ENTIRELY** based on test score + capstone quality.
6. Use YouTube, official documentation, Kaggle, and trusted internet resources to learn.
7. If stuck, ask your mentor — don't waste hours in silence. Communication matters.

03 PREREQUISITE TECHNICAL SKILLS

Master all topics below using YouTube tutorials, official documentation, and hands-on practice. Every topic listed will be tested in the Prerequisite Test and/or the Capstone Project.

A. Python Programming

Topic	Key Concepts	Resources
Python Basics	Variables, data types, loops, functions, file I/O, error handling, modules	YouTube: Corey Schafer, CS Dojo docs.python.org
NumPy	Arrays, indexing, slicing, broadcasting, math operations, random, reshaping	YouTube: Keith Galli numpy.org/doc
Pandas	DataFrame, Series, read_csv, groupby, merge/join, pivot, apply, loc/iloc, fillna	YouTube: Corey Schafer Pandas Series pandas.pydata.org
Matplotlib & Seaborn	Line/bar/scatter/heatmap plots, styling, subplots, sns themes	YouTube: sentdex, Kite matplotlib.org
Flask	Routes, request/response, templates, REST APIs, JSON endpoints, forms	YouTube: Traversy Media, Tech With Tim flask.palletsprojects.com
Jupyter Notebook	Cells, markdown, magic commands, exporting, widgets	YouTube: Corey Schafer jupyter.org

B. SQL & PostgreSQL

Topic	Key Concepts	Resources
SQL Fundamentals	SELECT, WHERE, ORDER BY, GROUP BY, HAVING, LIMIT, DISTINCT, aliases	YouTube: freeCodeCamp SQL Full Course w3schools.com/sql
Joins & Subqueries	INNER, LEFT, RIGHT, FULL JOIN, subqueries, CTEs (WITH clause)	YouTube: Socratica, Alex the Analyst postgresqltutorial.com
Aggregations & Window Functions	COUNT, SUM, AVG, MIN, MAX, ROW_NUMBER, RANK, LAG, LEAD, PARTITION BY	YouTube: Alex the Analyst postgresqltutorial.com
PostgreSQL Specific	pgAdmin setup, psycopg2 (Python connector), data types, SERIAL, JSONB, indexes	YouTube: Traversy Media PostgreSQL postgresql.org/docs
Data Cleaning in SQL	CAST, COALESCE, NULLIF, string functions, date functions, CASE WHEN	YouTube: Data with Mo kaggle.com/learn/advanced-sql

03 PREREQUISITE SKILLS (CONTINUED)

C. Data Analysis & Exploratory Data Analysis (EDA)

Topic	Key Concepts	Resources
EDA Process	Understand data shape, distributions, missing values, outliers, correlations	YouTube: Ken Jee, Rob Mulla Kaggle EDA notebooks
Statistics	Mean, median, mode, std deviation, variance, percentiles, Z-score, distributions	YouTube: StatQuest (Josh Starmer) khan academy statistics
Data Cleaning	Handle missing values, duplicate removal, type conversion, standardization, encoding	YouTube: Keith Galli pandas docs
Feature Engineering	Creating new features, binning, encoding categorical variables, normalization	YouTube: Krish Naik scikit-learn user guide
Visualization for Insights	Choosing the right chart, storytelling with data, dashboards, annotations	YouTube: Data Professor storytelling with data (book)

D. Machine Learning Fundamentals (Awareness Level)

Topic	Key Concepts	Resources
ML Concepts	Supervised vs unsupervised, training/testing, overfitting, underfitting, cross-validation	YouTube: StatQuest, Krish Naik scikit-learn.org
Regression	Linear regression, polynomial regression, R-squared, RMSE, MAE	YouTube: Andrew Ng (Coursera preview) Kaggle Learn
Classification	Logistic regression, decision trees, confusion matrix, precision, recall, F1	YouTube: Sentdex, StatQuest Kaggle Learn
scikit-learn	fit(), predict(), train_test_split, pipeline, preprocessing, model evaluation	YouTube: Corey Schafer sklearn scikit-learn.org/stable

E. Git & Version Control

Topic	Key Concepts	Resources
Git Basics	git init, clone, add, commit, push, pull, branch, merge, .gitignore, GitHub repo	YouTube: Traversy Media Git Crash Course git-scm.com/docs
Collaboration	Pull requests, code review, resolving conflicts, README writing	YouTube: freeCodeCamp Git & GitHub github.com guides

04 FINTECH DOMAIN KNOWLEDGE (MANDATORY)

As our company operates in the Fintech domain, domain knowledge is NON-NEGOTIABLE. You will be expected to apply data analytics to financial data. Learn the following thoroughly.

A. Stock Market Fundamentals

Concept	What You Must Know	Where to Learn
Stock Market Basics	What is a stock, equity, stock exchange, IPO, NSE/BSE (India), NYSE/NASDAQ, India website	YouTube: CA Rachana Ranade
Price & Volume	OHLC (Open, High, Low, Close), market trading volume, bid-ask spread, market orders	YouTube: Pranjal Kamra Zerodha Varsity (free)
Indices	NIFTY 50, SENSEX, sectoral indices, how index is calculated, index funds	YouTube: Groww Official zerodha.com/varsity
Technical Analysis (Basics)	Candlestick charts, support/resistance, MA, EMA, RSI, MACD, Bollinger Bands	YouTube: Nitin Bhatia TA Series Zerodha Varsity TA module
Fundamental Analysis (Basics)	P/E ratio, EPS, P/B ratio, revenue, profit, annual report reading, balance sheet	YouTube: CA Rachana Ranade TickerTape, Screener.in
Market Data APIs	Fetching stock data using Python: yfinance, NSEpy, Alpha Vantage API	YouTube: NeuralNine stock data finance.yahoo.com

B. Mutual Funds

Concept	What You Must Know	Where to Learn
Mutual Fund Basics	What is a MF, NAV, units, AUM, AMC, fund manager, expense ratio, exit load	YouTube: CA Rachana Ranade MF Series AMFI India website (amfiindia.com)
Types of Funds	Equity, debt, hybrid, ELSS, index funds, liquid funds, sectoral funds, FOF	YouTube: Groww Official ValueResearchOnline
Returns & Risk	CAGR, absolute return, XIRR, rolling returns, Sharpe ratio, alpha, beta, standard deviation	YouTube: Pranjal Kamra Motilal Oswal
SIP & Lump Sum	Systematic Investment Plan, rupee cost averaging, lump sum vs SIP comparison	YouTube: Yadnya Investment Academy ET Money SIP Calculator
Fund Selection	Star ratings, portfolio overlap, category benchmark comparison, fund house reputation	YouTube: NRI Money Clinic ValueResearchOnline, Kuvera

04 FINTECH DOMAIN KNOWLEDGE (CONTINUED)

C. Additional Fintech Concepts

Concept	What You Must Know	Where to Learn
Personal Finance	Budgeting, savings rate, net worth, debt management, financial planning	YouTube: Labour Law Advisor ET Money Blog
Banking & Payments	UPI, NEFT, RTGS, IMPS, payment gateway, digital wallets, fintech disruption	YouTube: CRED Curiosity Room RBI website (rbi.org.in)
Insurance Basics	Term, health, ULIP, premium, sum assured, claim ratio, LIC vs private	YouTube: CA Rachana Ranade PolicyBazaar blog
Taxation Basics	Income tax slabs, TDS, capital gains tax (STCG/LTCG), 80C deductions	YouTube: Yadnya, CA Rachana Ranade Incometaxindia.gov.in
Fintech Ecosystem	Neobanks, robo-advisors, lending tech, insurtech, regtech, BNPL, crypto basics	YouTube: Finshots finshots.in (newsletter)
Regulatory Bodies	SEBI, RBI, IRDAI, AMFI, their roles and how they regulate fintech	SEBI.gov.in, RBI.org.in YouTube: IPO corner

D. Essential Tools & Environments

Tool	Purpose	Learn From
VS Code	Primary IDE for Python development, extensions, debugging, Git integration	YouTube: Traversy Media VS Code code.visualstudio.com/docs
Jupyter Notebook / JupyterLab	Interactive data analysis, EDA, ML experiments, data storytelling	YouTube: Corey Schafer Jupyter jupyter.org/documentation
pgAdmin 4	PostgreSQL GUI, writing queries, table management, data import/export	YouTube: Traversy Media PostgreSQL pgadmin.org/docs
Google Colab	Free cloud Jupyter, GPU access, collaboration, sharing notebooks	colab.research.google.com YouTube: Data Professor Colab
Postman	API testing, REST endpoint testing, JSON response inspection	YouTube: Traversy Media Postman postman.com/learning-center
Excel / Google Sheets	Quick data inspection, pivot tables, basic charts, VLOOKUP, data cleaning	YouTube: ExcellsFun support.google.com/docs

05 WEEK 1 DAILY LEARNING PLAN

Day 1: Orientation & Python Foundation

- Attend orientation briefing. Read this entire document.
- Python: Variables, data types, lists, dicts, loops, functions
- Python: OOP basics (classes, objects, inheritance)
- Setup: Install Python, VS Code, Jupyter, pgAdmin
- Target: Write a Python script that reads a CSV and prints stats

Day 2: NumPy, Pandas & Data Cleaning

- NumPy: Arrays, operations, broadcasting, indexing
- Pandas: DataFrames, read_csv, head/tail, info, describe
- Pandas: loc/iloc, filtering, groupby, merge, fillna
- Practice: Load a dataset from Kaggle, clean it, explore it
- Target: Complete 1 EDA notebook on any financial dataset

Day 3: SQL + PostgreSQL + Capstone Assigned

- SQL: SELECT, WHERE, GROUP BY, HAVING, JOINS
- PostgreSQL: Setup pgAdmin, create DB, import CSV data
- Advanced SQL: CTEs, Window Functions, CASE WHEN
- CAPSTONE PROJECT IS ASSIGNED TODAY — Start planning immediately
- Target: Run 10+ SQL queries on financial data in PostgreSQL

Day 4: Visualization + Flask + Capstone Work

- Matplotlib & Seaborn: Create 5+ chart types with financial data
- Flask: Build a simple REST API with 2-3 endpoints
- Flask: Connect to PostgreSQL using psycopg2, return JSON
- Capstone: Write project structure, begin data analysis
- Target: Flask app running + first capstone data insights

Day 5: Fintech Domain + ML Basics + Capstone

- Stock market: OHLC data, fetch using yfinance in Python
- Mutual funds: NAV data, CAGR calculation in Python/Pandas
- ML Basics: Linear regression on a financial dataset
- Capstone: Complete data analysis section, begin modeling
- Target: Fintech domain notes ready. Capstone 50%+ complete

06 WEEK 2: TEST & CAPSTONE SUBMISSION

Online Prerequisite Test

The test will be conducted online during Week 2 (Days 6-7). It covers ALL technical and domain topics from this document. Format includes MCQs, short code outputs, SQL queries, and Fintech concept questions.

Test Section	Topics Covered	Weightage
Python & Libraries	NumPy, Pandas, Matplotlib, Flask, OOP	30%
SQL & PostgreSQL	Queries, Joins, Window Functions, pgAdmin	25%
Data Analysis & EDA	Statistics, data cleaning, visualization	20%
Fintech Domain	Stock market, MF, SEBI, financial ratios	15%
ML & Tools	sklearn basics, Git, Jupyter, Colab	10%

Capstone Project Guidelines

The Capstone Project is a real-world data analytics project assigned to you in Week 1. It will be in the Fintech domain. You must submit a complete notebook + Flask API + report.

- ✓ Exploratory Data Analysis (EDA) with visualizations on the provided financial dataset
- ✓ Data cleaning pipeline using Pandas (handle missing values, outliers, data types)
- ✓ SQL queries in PostgreSQL to answer 5+ business questions from the dataset
- ✓ Statistical insights: trends, correlations, anomaly detection
- ✓ A simple Flask API (minimum 2 endpoints) that serves data insights as JSON
- ✓ Optional (bonus): A basic ML model (regression or classification) on the dataset
- ✓ Final report: PDF or Jupyter Notebook with narrative, charts, and findings
- ✓ GitHub repository with clean commit history, README, and requirements.txt

07 POST-INTERNSHIP: PROJECT ASSIGNMENT CRITERIA

After Week 2, your mentor team evaluates your test and capstone and assigns you to ongoing company projects in the Fintech domain. The better you perform, the more critical the project you get.

Performance Band	Test Score	Capstone Quality	Project Assigned
Outstanding	90%+	Exceptional — production ready	Lead Analytics on Live Fintech Product
Excellent	75–90%	Strong — thorough analysis	Core Data Pipeline / Dashboarding Project
Good	60–75%	Good — solid foundations shown	Supporting role in Analytics Project
Needs Improvement	<60%	Incomplete or minimal work	Extended learning period, re-evaluation

08 LEARNING RESOURCES QUICK REFERENCE

Resource	URL / Platform	Best For
Zerodha Varsity	zerodha.com/varsity	Stock market & MF — free, excellent
Kaggle Learn	kaggle.com/learn	Python, Pandas, SQL, ML — free with certs
freeCodeCamp YouTube	youtube.com/@freecodecamp	Full-length coding courses
StatQuest (YouTube)	youtube.com/@statquest	Statistics, ML explained simply
Alex the Analyst	youtube.com/@alextheanalyst	SQL, Pandas, Power BI for data analysts
CA Rachana Ranade	youtube.com/@CARachanaRanade	Indian stock market & MF (Hindi/English)
PostgreSQL Tutorial	postgresqtutorial.com	Complete PostgreSQL + psycopg2 guide
Real Python	realpython.com	Python, Flask, pandas deep dives
Towards Data Science	towardsdatascience.com	EDA, ML, Fintech data science articles
NSE India / BSE	nseindia.com / bseindia.com	Real market data, historical prices
yfinance Docs	pypi.org/project/yfinance	Fetch stock data in Python
Flask Docs	flask.palletsprojects.com	Official Flask documentation
NeuralNine (YouTube)	youtube.com/@NeuralNine	Flask, data science, Python projects

09 QUICK CHECKLIST BEFORE WEEK 2 TEST

Python	■ Variables, loops, functions, OOP ■ NumPy arrays + operations ■ Pandas: read, clean, group, merge ■ Matplotlib/Seaborn: 5+ chart types ■ Flask REST API with JSON responses
SQL	■ SELECT, WHERE, GROUP BY, HAVING ■ INNER, LEFT, RIGHT JOINs ■ CTEs and Window Functions ■ PostgreSQL setup + psycopg2 ■ 5+ business queries written
Fintech	■ Stock: OHLC, indices, P/E, EPS ■ Mutual Fund: NAV, CAGR, SIP, expense ratio ■ yfinance to fetch stock data ■ SEBI/RBI/AMFI roles understood ■ At least 1 financial dataset analyzed

Tools

- VS Code + Jupyter set up
- pgAdmin connected to local DB
- Git repo created with commits
- Google Colab account active

Good luck! The internship is your launchpad — put in the effort this week and you will walk away with real skills, a strong portfolio project, and a company project to your name.